Research, Engineering & Innovation Portfolio

Ferdaous MASMOUDI

PhD in Electrical Engineering

Postdoctoral Researcher | R&D Director | Technology Entrepreneur

Mechatronics • Robotics • Embedded Systems & IoT
Control & Instrumentation • Intelligent Systems • Industrial Automation
Renewable Energy • Power Electronics • Energy Harvesting & Storage

Sfax, Tunisia

**I build engineering systems,
and I teach students
how to build them.**

*Bridging research,
technology development,
and real-world implementation.*

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Online Research Portfolio

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https://ferdaousmasmoudi.github.io/

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Ferdaous Masmoudi

Electrical Engineer | Researcher | R&D & Innovation

Professional Profile

Electrical engineer with a PhD in Electrical Engineering, an MSc in Electrical Conversion and Renewable Energy, and current Post-Doctoral research experience within the INTERREG NEXT Italy–Tunisia FLOTTANT project. Combines academic research, university teaching, and approximately ten years of applied R&D and engineering experience.

Her work spans mechatronics, robotics, embedded systems and IoT, control and instrumentation, renewable energy, power electronics, energy storage, and intelligent engineering systems, with experience ranging from mathematical modeling and simulation to embedded implementation, prototyping, experimentation, and system validation.

Current activities focus on intelligent and autonomous engineering systems, hybrid energy harvesting, multiphysics system integration, embedded intelligence, and the translation of research concepts into practical engineering solutions for industrial, agricultural, energy, and environmental applications.

Current Positions

2026–Present **Postdoctoral Researcher**, *FLOTTANT – INTERREG NEXT Italy–Tunisia*, National Engineering School of Sfax (ENIS), University of Sfax
Hybrid marine energy harvesting, autonomous instrumentation, multiphysics modeling, system identification, and experimental validation.

2023–Present **Co-Founder & R&D Director**, *QUANTITECK*, Sfax, Tunisia
Research, engineering, and technology development in robotics, intelligent embedded systems, industrial IoT, and smart engineering solutions.

Education and Postdoctoral Training

2026–Present **Postdoctoral Researcher**, *FLOTTANT Project – INTERREG NEXT Italy–Tunisia, National Engineering School of Sfax (ENIS), University of Sfax*
Research on hybrid marine energy harvesting, autonomous instrumentation, multiphysics system integration, energy management, and experimental validation.

2016 **PhD in Electrical Engineering**, *National Engineering School of Sfax (ENIS), University of Sfax*, Sfax, Tunisia
Dissertation: *Optimization of Standalone Photovoltaic Systems*.
Research in photovoltaic system modeling, parameter identification, optimization, solar tracking robotic system, MPPT control, electrochemical energy storage, power electronics, and experimental validation.

2012 **MSc in Electrical Conversion and Renewable Energy**, *National Engineering School of Sfax (ENIS), University of Sfax*, Sfax, Tunisia
Master's thesis: *Modeling and MPPT Control of a Photovoltaic Conversion System*.
Research in photovoltaic generator modeling, DC–DC power conversion, maximum power point tracking, simulation, and control.

2011 **Engineering Degree in Electrical Engineering**, *National Engineering School of Sfax (ENIS), University of Sfax*, Sfax, Tunisia
Core training in electrotechnics, electrical machines, power systems, industrial automation, control, power electronics, and instrumentation.

2006–2008 **Preparatory Cycle for Engineering Studies**, *Preparatory Institute for Engineering Studies of El Manar*, Tunis, Tunisia
Intensive university-level training in mathematics, physics, and engineering sciences.

2006 **Baccalaureate in Mathematics, with Honors**, *Pioneer High School of Ariana (Lycée Pilote Ariana, LPA)*, Ariana, Tunisia

Industrial R&D and Entrepreneurship

2023–Present **Co-founder and R&D Director**, *QUANTITECH*, Sfax, Tunisia

- Co-founded a technology company focused on robotics, intelligent embedded systems, industrial IoT, and digital solutions for industrial and agricultural applications.
- Leading the system-level design and development of PLUMA, an intelligent mobile robotic platform for poultry-farm monitoring and assistance.
- Defining and coordinating the integration of locomotion, embedded control, environmental sensing, robotic perception, communication, data acquisition, remote supervision, and cloud connectivity.
- Developing industrial and AgriTech solutions for autonomous monitoring, environmental sensing, traceability, process digitalization, and decision support.
- Contributing to product definition, technical planning, prototype development, experimental testing, industrial partnerships, and technology-transfer activities.

2016–2023 **Co-founder and R&D Lead**, *NOVEL-TI*, Sfax, Tunisia

- Directed applied research and system development in intelligent energy systems, embedded architectures, mobile robotics, instrumentation, and AI-enabled IoT platforms.
- Designed and supervised the end-to-end development of hardware–software systems, including photovoltaic-powered platforms, mobile robotic systems, multi-sensor acquisition architectures, and cloud-connected supervision dashboards.
- Led the development of AgriTech and industrial digitalization solutions integrating LoRa and Wi-Fi communication, RFID/NFC data logging, QR-based identification, and blockchain-enabled traceability using Hyperledger Fabric.
- Supervised multidisciplinary engineering and end-of-study projects covering system requirements, architecture design, electronics, firmware, software integration, prototyping, testing, troubleshooting, and technical validation.
- Coordinated technical teams and international collaborations with European industrial partners, from initial system specification to prototype delivery and field-oriented evaluation.

University Teaching Experience

2019–2021 **Lecturer in Mechatronics**, *Private Institute of Applied Sciences (IPSAS)*, Sfax, Tunisia
Delivered lectures, tutorials (TD), and practical/laboratory sessions (TP) in Mechatronics across several academic years.

- **Course scope:** Mechatronic system design; mechanical structures and actuators; electrical and electronic systems; embedded systems; sensors and data acquisition; control and automation; software integration.
- **Robotics and prototyping:** Microcontrollers, Arduino, Raspberry Pi, STM32, PLCs, ROS, robotic arms, mobile robots, and autonomous systems.
- **Emerging technologies:** Introductory concepts in computer vision, artificial intelligence, IoT, wireless communication, and cloud-connected systems.
- **Teaching approach:** Problem-based and project-based learning through tutorials, system-design exercises, homework assignments, practical activities, and student projects.

2017–2018 **Contract Lecturer**, *Higher Institute of Industrial Management (ISGI)*, Sfax, Tunisia

Delivered two courses at Bachelor's and Professional Master's levels.

- **Maintenance of Mechatronic Systems** – Second-year Professional Master's programme in Management and Maintenance of Industrial Systems. Introduced the architecture and constituent disciplines of mechatronic systems, including mechanics, electrical and electronic systems, sensors and actuators, control, embedded systems, and software. Students were trained to decompose complex mechatronic equipment into functional subsystems for system analysis, diagnosis, and maintenance.
- **Open-Source Software Development** – Third-year Applied Bachelor's programme in Industrial Management Informatics. Introduced open-source software principles and practical development using Android, Python, and other open-source tools and environments, combining programming fundamentals with hands-on application development and implementation.
- **Assessment:** Responsible for continuous assessment and final examination grading.

2017–2018 **Contract Lecturer / Laboratory Instructor**, *Faculty of Sciences of Sfax (FSS), University of Sfax*, Sfax, Tunisia

Delivered practical/laboratory teaching in **Instrumentation Computing (*Informatique d'Instrumentation*)** for second-year Applied Bachelor's students in Physics.

- **Laboratory content:** LabVIEW programming, virtual instrumentation, graphical programming, computer-based data acquisition, signal processing, visualization, and experimental instrumentation.
- **Practical tools and methods:** Sub-VIs, control structures, data visualization, NI-DAQ, and Measurement & Automation Explorer (MAX) for acquisition, device configuration, and experimental testing.
- **Assessment:** Responsible for laboratory-work assessment and grading.

Teaching Portfolio

Electrical Engineering Foundations Electric Circuits; Analog and Digital Electronics; Electrotechnics; Electrical Machines; Signals and Systems; Electrical Measurements; Power Systems Fundamentals.

Control, Instrumentation & Automation Control Systems; Digital Control; Sensors and Instrumentation; Data Acquisition; Industrial Automation; PLC Programming; System Identification.

Mechatronics & Robotics Mechatronics; Mechatronic System Design; Sensors and Actuators; Mechatronic Maintenance; Robotics; Mobile Robotics; ROS; Motion Control and System Integration.

Embedded Systems & IoT Microcontrollers; Embedded Systems; Embedded C/C++; Python for Engineering; Industrial IoT; Edge Computing; Cyber-Physical Systems; Industrial Communication.

Energy & Power Electronics Power Electronics; DC–DC Converters; Renewable Energy Systems; Solar Photovoltaics; Energy Storage; Energy Harvesting; Energy Management.

Modeling & Simulation Dynamic System Modeling; MATLAB/Simulink; COMSOL Multiphysics; LabVIEW; Parameter Identification; Engineering Optimization; Experimental Validation.

Intelligent Systems & AI Computer Vision; Edge and Embedded AI; Intelligent Sensing; AI for Robotics; Autonomous Systems.

Teaching Approach Hands-on, student-centered teaching combining engineering fundamentals with modeling, design, implementation, experimentation, and validation through tutorials, laboratory work, multidisciplinary projects, and real-world engineering problems.

Selected Student Supervision and Applied Engineering Projects

2020 **PhD Co-Supervisor – Rabeb Abid**, *National Engineering School of Sfax (ENIS), University of Sfax*, Sfax, Tunisia

Thesis: *Design and Control of Static Converters for Standalone Photovoltaic Applications*.
Co-supervision of doctoral research on standalone photovoltaic conversion, DC–DC Luo converters, MPPT control, energy storage, power electronics, and experimental validation.

2022 **Engineering Final-Year Project Supervisor – Younes Rayen Chabchoub**, *National School of Electronics and Telecommunications of Sfax (ENET'COM)*, Sfax, Tunisia
Project: *Autonomous Weather Station*.
Supervision of the design and implementation of an autonomous agricultural monitoring system integrating environmental sensors, ESP32-based embedded electronics, LoRa/LoRaWAN communication, remote data transmission, and connected monitoring.

2025 **Engineering Final-Year Project Supervisor – Seif Ben Ali**, *National School of Computer Science (ENSI), University of Manouba*, Tunisia
Project: *AgriTrace – Blockchain-Based Food Traceability*.
Professional supervision of a food-traceability proof of concept integrating Hyperledger Fabric, IPFS, smart contracts, role-based access, QR-based identification, distributed storage, web technologies, testing, and system validation.

2025 **Final-Year Project Co-Supervisor – Asma Trabelsi and Yosser Soussou**, *National School of Electronics and Telecommunications of Sfax (ENET'COM)*, Sfax, Tunisia
Project: *RoboLinker – Mobile-Sensor-Based Robotic Control*.
Co-supervision of a robotic control and monitoring platform integrating a mobile robot, Android smartphone, embedded environmental sensors, Firebase cloud services, robot localization, real-time monitoring, and embedded AI for poultry detection.

Research Vision

My research aims to develop intelligent, autonomous, and energy-aware engineering systems by integrating mechatronics, embedded AI, model-based design and control, and autonomous energy technologies.
Future work will particularly focus on robotic and sensing platforms capable of perceiving their environment, managing their energy resources, adapting their behavior, and operating reliably in industrial, agricultural, and environmental applications.

Research Projects and Contributions

Hybrid Marine Energy Harvesting & Autonomous Instrumentation Postdoctoral research within FLOTTANT, an INTERREG NEXT Italy–Tunisia project, addressing hybrid marine energy harvesting and autonomous instrumentation. Current contributions include experimental characterization, signal processing, reduced-order and grey-box modeling, genetic-algorithm-based parameter identification, and validation of piezoelectric cable transducers. The broader project integrates piezoelectric, capacitive, thermoelectric, and microbial energy-conversion technologies within a floating marine instrumentation platform.

Photovoltaic Systems, Energy Conversion & Storage Doctoral and subsequent research on standalone photovoltaic systems, including photovoltaic generator modeling, DC–DC power conversion, MPPT control, solar tracking, electrochemical energy storage, battery state estimation, energy management, and experimental validation.

Modeling, Identification & Control Development of nonlinear, state-space, grey-box, and data-assisted models for photovoltaic and electrochemical systems. Contributions include parameter identification, state estimation, least-squares and ARX approaches, Kalman filtering, optimization, genetic algorithms, and control-oriented experimental validation.

Embedded & Experimental Engineering Translation of mathematical and simulation models into experimental systems through MATLAB/Simulink, C++ and Python implementation, microcontroller and SoC platforms, power-electronic converters, motor-control architectures, multi-sensor acquisition, signal conditioning, real-time processing, and experimental testing.

Research Internships and Industrial Training

2013–2014 **LSIS Laboratory, Polytech Marseille, Marseille, France** , *Research Internship – Battery Observation and Management System (BOMS)*

- Contributed to the modeling and analysis of electrochemical energy-storage systems for battery monitoring and energy-management applications.
- Investigated parameter estimation, state observation, and state-of-charge assessment for control-oriented battery supervision.
- Developed and evaluated simulation models using MATLAB/Simulink, followed by embedded implementation on microcontroller-based platforms.
- Participated in laboratory testing, experimental-data analysis, and validation of battery monitoring and management algorithms.

2012 **Chemnitz University of Technology, Chemnitz, Germany**, *Research Internship – Residential Photovoltaic System Integration*

- Studied the integration of photovoltaic generation systems into residential electrical environments.
- Conducted component-level modeling and simulation of photovoltaic energy-conversion architectures.
- Evaluated system configurations and operating strategies to support the sizing and optimization of residential photovoltaic installations.
- Strengthened international research experience in renewable-energy modeling, simulation, and system integration.

2009–2011 **Industrial Training – Electrical Energy and Power Systems**, *STEG, OACA – Tunis-Carthage Airport, and SNDP AGIL*, Tunisia

- Gained practical exposure to national electrical-grid operation, energy dispatching, electrical distribution, and power-system supervision at the Tunisian Company of Electricity and Gas (STEG).
- Studied electrical installations, power distribution, protection, and operational requirements within the infrastructure of Tunis-Carthage International Airport.
- Observed industrial electrical equipment, medium-voltage systems, switchgear, maintenance procedures, and energy-management practices.
- Contributed to technical studies related to electrical installations and photovoltaic system sizing for industrial and commercial applications.

Technical Expertise

Modeling, Identification & Control	Nonlinear and grey-box modeling, state-space methods, parameter identification, state estimation, optimization, genetic algorithms, MPPT and model-based control	Embedded Systems & Edge Computing	STM32, ESP32, Arduino, NVIDIA Jetson, Raspberry Pi, Odroid, embedded C/C++, firmware development, I2C, SPI, UART
Simulation, Multiphysics & Instrumentation	MATLAB/Simulink, COMSOL Multiphysics, LabVIEW, data acquisition, signal processing, sensor interfacing, experimental instrumentation and validation	Robotics & Autonomous Systems	ROS/ROS 2, Gazebo, RViz, mobile robotics, sensor–actuator integration, embedded motion control, robotic perception
Electronics & Hardware Design	Altium Designer, KiCad, EasyEDA, Proteus ISIS/ARES, PCB design, electronic prototyping, motor-driver and sensor integration	Energy & Power Electronics	Solar PV systems, MPPT, DC–DC converters, electrochemical energy storage, energy harvesting, power management
Industrial IoT & Connectivity	MQTT, LoRa/LoRaWAN, Zigbee, Wi-Fi, BLE, GSM, Modbus, RFID/NFC, connected sensing and remote monitoring	AI & Computer Vision	OpenCV, YOLO, TensorFlow Lite, ONNX, edge AI, embedded inference and computer vision
Software Development	Python, C++, MATLAB, Bash, GNU Make, Git, Linux development environments	Mobile, Cloud & Connected Systems	Android, REST APIs, Firebase, cloud synchronization, remote supervision, data logging and connected applications

System Integration & Prototyping	Hardware–software co-design, multi-sensor integration, prototype development, debugging, troubleshooting, testing and experimental validation	Emerging AI Systems	Agentic AI for autonomous engineering systems (current development focus)

Languages

Arabic: Native • French: Fluent • English: Professional proficiency • German: Basic knowledge

Publications

Peer-Reviewed Journal Articles

2021 Loukil, J., **Masmoudi, F.**, and Derbel, N., "A real-time estimator for model parameters and state-of-charge of lead-acid batteries in photovoltaic applications," *Journal of Energy Storage*, Elsevier, 2021.

2019 Jarraya, I., **Masmoudi, F.**, et al., "An online state-of-charge estimation for lithium-ion and supercapacitor in hybrid electric drive vehicle," *Journal of Energy Storage*, Elsevier, 2019.

2014 **Masmoudi, F.**, "Identification of internal parameters of a mono-crystalline photovoltaic cell model with experimental validation," *International Journal of Renewable Energy Research*, 2014.

Book Chapters

2020 Loukil, J., **Masmoudi, F.**, and Derbel, N., "Third order model and identification of lead-acid batteries using meta-heuristic algorithms and experimental measurements," Springer book chapter, 2020.

2019 Boujelben, N., **Masmoudi, F.**, Djemel, M., and Derbel, N., "Modeling and comparison of boost converter with cascaded boost converters," *Green Energy and Technology*, Springer, 2019.

Peer-Reviewed Conference Papers

2019
- **IEEE SSD** – "MPPT control strategies for photovoltaic applications: algorithms and comparative analysis."

2018
- **IEEE SSD** – "Modeling and parameters estimation for lithium-ion cells in electric drive vehicle."
- **IEEE SSD** – "Performance analysis of Luo-converter for photovoltaic application."

2017
- **IEEE SSD** – "Design and comparison of quadratic boost and double cascade boost converters with boost converter."
- **IEEE SSD** – "Comparative study of fundamental variable-input converters used for photovoltaic conversion systems."
- **IEEE SSD** – "State-of-charge estimation of lead-acid battery using a Kalman filter."
- **SM2C** – "Comparative study of the performances of the DC/DC Luo-converter in photovoltaic applications."

2016
- **IREC** – "Modeling and simulation of conventional DC–DC converters dedicated to photovoltaic applications."
- **IEEE SSD** – "Single and double diode models for conventional monocrystalline solar cells with extraction of internal parameters."
- **IEEE SSD** – "Modeling of internal parameters of a lead-acid battery with experimental validation."

2015
- **STA** – "Design and realization of a photovoltaic system controlled by an MPPT algorithm."

2014
- **SAAEI, Tangier, Morocco** – "Simple models design for one monocrystalline photovoltaic cell by identification of internal parameters."

2012 ○ **EVER, Monaco** – "Oil service station PV-based power supply: a challenging Tunisian national program."

Academic Engagement and Professional Development

Conference Activities: Participation in the IEEE International Conference on Systems, Signals and Devices (SSD), including scientific communication and engagement with research activities in electrical engineering, control, energy systems, and embedded technologies.

Technical Workshops: Participation in technical workshops and professional development activities related to IoT, STM32 microcontrollers, Arduino platforms, embedded systems, and intelligent engineering applications.

Industry–Academia Engagement: Contribution to collaborative activities connecting university teaching, student engineering projects, applied R&D, prototype development, and industrial problem solving.

References

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Postdoctoral Supervisor: **Prof. Nabil Derbel**, *Full Professor, Department of Electrical Engineering*, Control and Energy Management Laboratory (CEM Lab), ENIS, Sfax, Tunisia
Email: nabil.derbel@enis.tn

Academic Collaborator: **Dr. Imen Jarraya**, *Postdoctoral Researcher*, Robotics and Internet-of-Things Laboratory (RIOTU Lab), Prince Sultan University, Riyadh, Saudi Arabia
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